Physics of the Earth and Planetary Interiors, 52 (1988) 256–266
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeomagnetic evidence of tectonic rotations in the southern margin of the Inner West Carpathians

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(Received May 29, 1987; revision accepted September 26, 1987)

Márton, E., Márton, P. and Less, G., 1988. Palaeomagnetic evidence of tectonic rotations in the southern margin of the Inner West Carpathians. Phys. Earth Planet. Inter., 52: 256–266.

Palaeomagnetic samples were collected at 33 Triassic localities from three nappes and a pseudo-autochthonous unit in the Aggtelek–Rudabánya Mountains in order to establish the rotation pattern of the area as a whole with respect to the enclosing principal continents, and to check whether relative movements in the past between the various units that make up the region are resolvable by the palaeomagnetic data.

After detailed thermal demagnetization and careful data selection it has emerged that the entire study area was rotated anticlockwise with respect to both Europe and Africa, i.e., in a similar manner and to a similar degree as other units in the Central Mediterranean region (e.g., the Transdanubian Central Mountains).

On the other hand, small-scale relative movements such as those between the Aggtelek and Rudabánya areas are not apparent in the present data.

1. Introduction

Reconnaissance palaeomagnetic measurements of Triassic rocks from the Aggtelek Mountains in the southernmost part of the Inner West Carpathians (IWCs) have indicated that the declination rotation in the Silica nappe is anticlockwise (Márton et al., 1981). Apparently, this result is in disagreement with earlier ones from different nappes of the IWCs of Slovakia, which have been interpreted (Krs, 1966; Krs et al., 1982) as indicators of clockwise rotations by variable angles relative to stable Europe. It is to be noted, however, that Muska and Vozár (1984) have recently made a reappraisal of some formerly studied localities and concluded that the rotation of the autochthonous part of the Slovakian Ore Mountains is anticlockwise.

Indeed, the problem of deciphering tectonic rotations in orogenic belts is difficult: first, because the rotations themselves can be very complex, and second, and perhaps more importantly, because the tectonic deformations may alter the primary magnetic signal to such an extent that what we observe now is just an overprint or the resultant of the primary (if any) and secondary remanences.

In the following, we present the palaeomagnetic results from an area which belongs to the southernmost extension of the IWCs. We have carried out this study with the aim of delineating the tectonic rotations both within the region itself and relative to the main continental masses.

2. Geology and sampling

The study area (Fig. 1) comprises two tectonically distinct areas, the Aggtelek and the Rudabánya Mountains. The former is a direct

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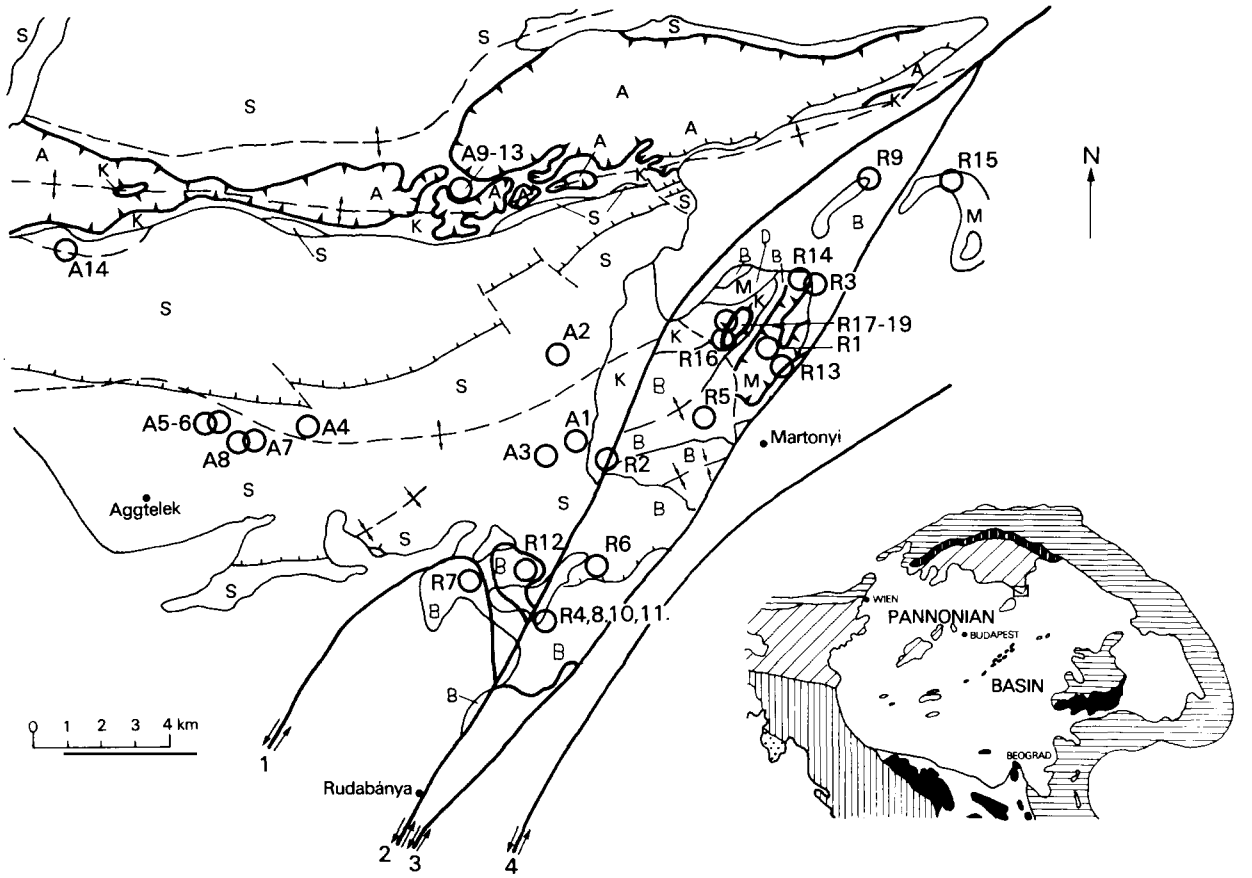


Fig. 1. Geological (tectonic) sketch map of the Aggtelek–Rudabánya Mountains with the sampling localities which are marked A1–A14 and R1–R19 in the Aggtelek and Rudabánya areas, respectively. The inset in the lower right corner shows the position of the study area in the Pannonian Basin. Features marked by 1–4 are left lateral faults constituting part of the Darnó fracture zone; 1 and 2 separate the Aggtelek (left) from the Rudabánya (right) Mountains. Other relevant symbols are as follows: S = Silica nappe, A = Alsó hegy nappe, K = Komjáti suite (friction breccia), B = Bódva nappe, M = Martonyi nappe, B = Bódvarákó pseudoautochthonous beds. Areas not marked are covered by Neogene and younger formations. The structural symbols applied are self-explanatory.

continuation of the IWCs, whereas the latter was brought into contact with it by left lateral movements along the Darnó fracture zone as late as Oligo-Miocene time.

The main mass of the Aggtelek Mountains is constituted by the Silica nappe, which consists of a neritic to pelagic sedimentary sequence of latest Permian to Middle Jurassic age. With a friction breccia in between the Silica nappe is underlain by the slightly metamorphosed Meliata Series, a different sedimentary suite which was formed in a deep oceanic environment during the Triassic and Jurassic periods.

The bulk of the Rudabánya Mountains is represented by the Bódva nappe, whose older beds can be correlated with those of the Silica nappe while in the younger ones a marked difference exists. To the north, the Bódva series is overlain by a slightly metamorphosed nappe fragment, the Martonyi nappe. At its base, again a friction breccia occurs, with the pseudo-autochthonous Bódvarákó Beds, whose rocks can partly be related to those of the Meliata Suite.

The two main nappes were emplaced in the Late Jurassic–Early Cretaceous. The folded and imbricated structure of the whole region dates

from the Middle Cretaceous. Subsequently, small-scale secondary nappes were formed in both areas (Alsóhegy nappe and Martonyi nappe). Finally, the Rudabánya Mountains became attached to the Aggtelek Mountains and within the latter the Alsóhegy nappe may have moved to some extent to the west.

For palaeomagnetism studies we sampled the Silica and Bódva nappes and two other units of the Rudabánya Mountains, the Martonyi nappe and the pseudo-autochthonous Bódvarákó Beds, collecting a total of 321 samples (cores) from 33 localities.

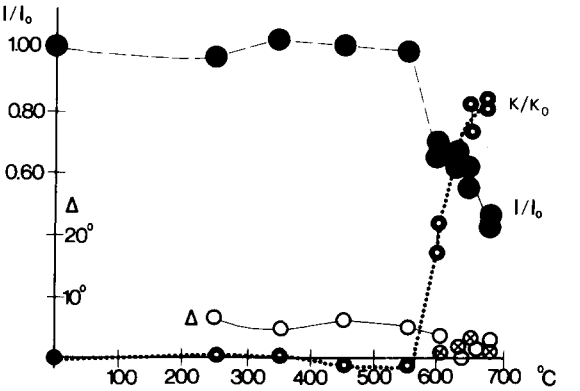


Fig. 2. Relative susceptibility (log scale), relative intensity and directional changes of NRM (linear scales) versus temperature, showing that in this sample (from locality A2) the NRM direction is not affected appreciably by the newly formed strongly magnetic phases after 550 ° C during demagnetization. k/k_0 = Relative susceptibility (o). I/I_0 = Relative intensity (●). △: Change in NRM direction between two consecutive demagnetization steps. First heating = ○. First and second heating in opposite orientation at the same temperature = ⊗.

3. Magnetization analysis

The natural remanence of each sample was analysed by stepwise thermal demagnetization aiming at total destruction of the remanence. Chemical changes induced by the thermal treatment were monitored by low field susceptibility measurement after each demagnetization step. This helped identify maghemite when the susceptibility decreased on heating. Usually, the susceptibility steeply increased at some elevated temperature owing to newly formed strongly magnetic phases. Whether or not this had affected the remanence measurement was checked by reheating the speci-

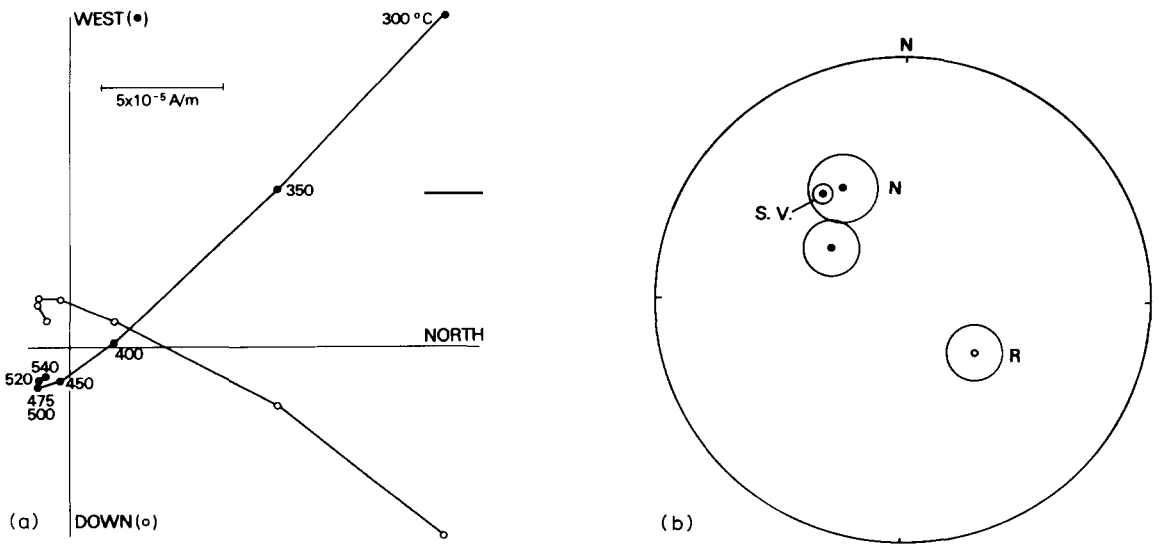


Fig. 3. (a) Orthogonal demagnetization diagram of a sample from locality A14. The initial susceptibility (-5×10^{-6} SI units) did not change during demagnetization. (b) Stereographic projection of mean remanence directions with α_{95} confidence circles from locality A14. N and R refer to the mean directions of the normal and reversed samples after 520 ° C. S.V., Mean direction of the subtracted remanences between 250 and 400 ° C. The unmarked mean direction corresponds to the inverse of R (● = positive inclinations, ○ = negative inclinations).

mens in opposite orientation at each temperature after the abrupt susceptibility increase had started. Figure 2 shows an example of the application of this method. The remanence measurements of the sample under consideration were repeatable within reasonable limits up to 680 ° C despite the susceptibility increase after 550 ° C, but there were less favourable cases, when the cleaning had to be stopped at the temperature of the onset of massive mineralogical alterations.

The characteristic remanences were isolated by studying both Zijderveld and stereographic plots as well as by computing subtracted remanence vectors when necessary. Typical examples are given in Figs. 3 and 4.

Figure 3a shows the behaviour of a sample from the Silica nappe (locality A14). On demagnetization, the remanence changes polarity from normal to reversed. The directions of the initial remanence and the subtracted vectors between 250 and 400 ° C are practically identical. From the 21 samples collected at this locality, 13 exhibited the same demagnetization behaviour, five changed neither polarity nor direction with respect to the initial values and the remaining three became too weak for precise measurement after 500 ° C. Figure 3b shows the mean directions with α_{95} values of the normal and reversed samples after 520 ° C and of the 21 subtracted vectors between 250 and 400 ° C. Obviously, the normal samples after 520 ° C are of the same direction as the subtracted remanences, whereas the reversed samples define a different direction after inversion to normal polarity. Because the initial remanence is usually a composite one, we conclude that only the reversed samples are free from overprinting after 520 ° C, hence their mean may represent the direction of the characteristic remanence for the locality in question (cf., Table I).

In Fig. 4a (locality R4) a high-temperature remanence is exemplified which persists between 500 and 625 ° C. The locality mean was computed from the directions of magnetization after 500 ° C (cf. Table II). The demagnetization diagram in Fig. 4b represents the behaviour of the samples from locality R9. Here, the tightest grouping was

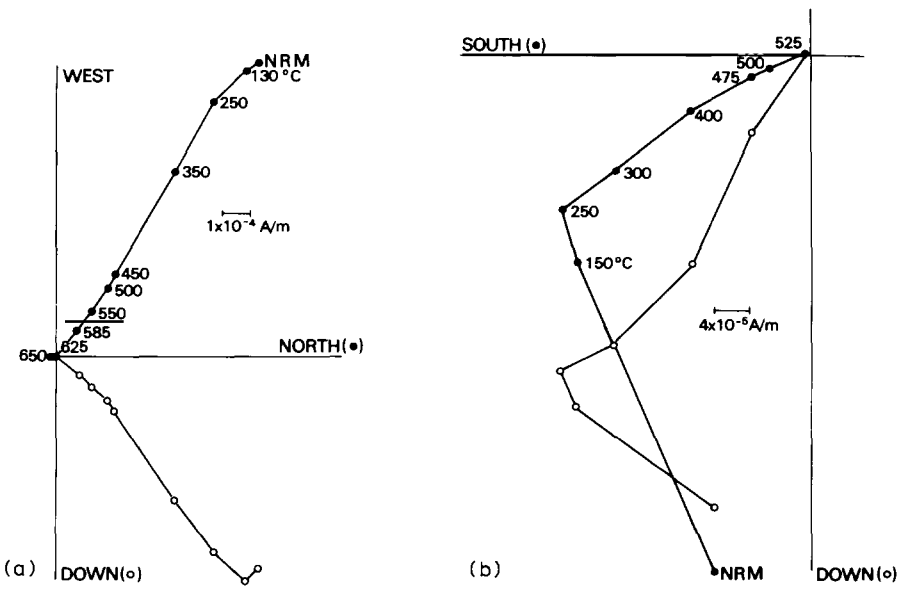


Fig. 4. (a) Orthogonal demagnetization diagram of a sample from locality R4. The initial susceptibility (2×10^{-6} SI units) increased by an order of magnitude on heating to 585 ° C and by a further half-order to 650 ° C but with no consequence on the intensity, which is not appreciable after 585 ° C. (b) Orthogonal demagnetization diagram of a sample from locality R9. The initial susceptibility (6×10^{-6} SI units) stayed constant up to 500 ° C, then increased by an order of magnitude at 525 ° C and the intensity dropped to a value near the noise level of the magnetometer (4×10^{-6} A m^{-1}).

TABLE I

Results of remanence measurements from the Aggtelek Mountains

Sampling locality	Rock	Geological age		Tectonic tilt dip (°)–bearing(°)	N/N_0	$D°$ $D°_c$	$I°$ $I°_c$	k	α_{95} (°)	Optimum cleaning temperature (°C)
		Stage	Epoch							
A1	Sandstone	Seisian	Early Triassic	9–233	6/6	105 103	−20 −14	19	17.6	670–680
A2*	Sandy marl	Early Campilian		16–237 5–265	6/10	256 255	20 10	49 32	9.7 12.0	525–600
A3	Marl	Early Campilian		27–270	6/6	304 292	54 31	24 26	16.9 13.4	500–525
A4	Limestone	Top of early triassic		57–323, 56–311 51–301	8/9	309 308	78 27	28 38	10.6 9.0	500–525
A5	Dolomite	Early Anisian	Middle Triassic	15–230	5/5	284 269	59 49	55	10.4	250–500
A6	Dolomite	Early Anisian		32–197	7/7	291 277	24 22	37	10.0	350
A7	Dolomite	Early Anisian		36–186	4/4	119 81	−44 −46	49	13.2	450–500
A8	Limestone	Late Anisian		21–192	6/6	300 282	40 43	15	18.2	400–500
A9	Limestone	Late Norian		35–91, 35–107 30–110	15/15	284 281	40 71	42 48	6.0 5.6	250–450
A10	Limestone	Late Norian		35–80	6/6	261 262	22 57	91	7.1	450
A11	Limestone	Late Norian	Late Triassic	35–14	4/4	243 298	52 61	45	13.7	450
A12	Limestone	Late Norian		44–48	4/4	259 305	36 65	83	10.1	500
A13	Limestone	Late Norian		35–0	3/3	247 281	41 45	137	10.6	500
A14	Limestone	Late Norian		38–314	13/21	58 126	−85 −51	11	12.9	520

A1–A14: Codes of sampling localities, (cf., Fig. 1). N: Number of useful samples. N_0: Number of samples collected. $D°$, $I°$ and $D_c°$, $I_c°$: mean declination and inclination before and after tectonic correction, respectively. k, α_{95}: Fisher's (1953) measures of precision and confidence of the mean.

* Results not considered suitable for palaeomagnetic (tectonic) interpretation (see section 3.1.1.).

achieved after 300 °C with $k = 27$ (cf. Table II), though after 400 °C $k = 24.4$ is almost as good. After 525 °C the intensities became weak and the directions scattered. Note the large difference between the directions of the thermally isolated remanences for R4 and R9 although the two localities belong to the same tectonic unit. This and other examples (cf., Table II) suggest that in our

TABLE II
Results of remanence measurements from the Rudabánya Mountains

Sampling locality	Rock	Geological age		Tectonic tilt dip (°)–bearing (°)	N/N_0	$D°$ $D°_c$	$I°$ $I°_c$	k	α_{95} (°)	Optimum cleaning temperature (°C)
		Stage	Epoch							
Bódva nappe (s.l.)										
R1 *	Dolomite	Early Anisian	Middle Triassic	22–35	8/9	305 317	31 28	59	7.3	300–400
R2 *	Dolomite	Early Anisian		38–132	0/10	Unstable magnetization				–
R3 *	Dolomite	Early Anisian		38–320	6/6	304 309	47 10	93	7.0	150–250
R4	Limestone	Middle Anisian		42–291	10/10	345 312	68 32	78	5.5	500
R5	Limestone	Late Anisian		77–230	6/10	25 277	31 61	47	9.9	400
R6	Limestone	Ladinian		64–284	12/12	136 278	81 33	55	5.9	400
R7	Limestone	Ladinian		40–334	8/8	265 302	62 35	38	9.1	250–500
R8	Limestone	Ladinian–Carnian		91–287	4/5	85 314	36 48	27	18.1	475–500
R9 *	Limestone	Ladinian–Carnian	Late Triassic	67–5	21/23	147 47	47 50	27	6.2	300
R10 *	Limestone	Carnian		87–301	15/15	208 292	81 4	42	6.0	400–500
R11 *	Limestone	Early Norian		73–294	12/12	230 287	83 15	108	4.2	250–500
R12 *	Limestone	Norian		25–40	10/12	35 39	83 58	41	7.6	525
Martonyi nappe										
R13 *	Limestone	Ladinian–Early Carnian	Middle–Late Triassic	133–298	5/11	4 49	−4 −15	44	11.7	350–450
R14 *	Limestone	Late Carnian	Triassic	50–328	11/14	303 305	29 −17	157	3.7	250
R15 *	Limestone	Late Norian		16–46 (170–226)	7/9	138 106 (166)	66 62 (−62)	58	8.0	400–515
Bódvarákó pseudo-autochthonous unit										
R16 *	Dolomite	Early Anisian	Middle Triassic	20–28	10/10	88 67	60 47	52	6.7	500
R17 *	Cherty Limestone	Anisian–Ladinian		25–24	15/18	333 9	80 58	26	7.7	400
R18 *	Limestone	Ladinian		27–72	6/6	2 47	76 56	427	3.2	300
R19 *	Aleurite	Ladinian		15–107	8/13	312 314	8 21	28	10.7	400

R1–R19: Codes of sampling localities/sites (cf., Fig. 1). Other symbols including the asterisks are the same as in Table I.

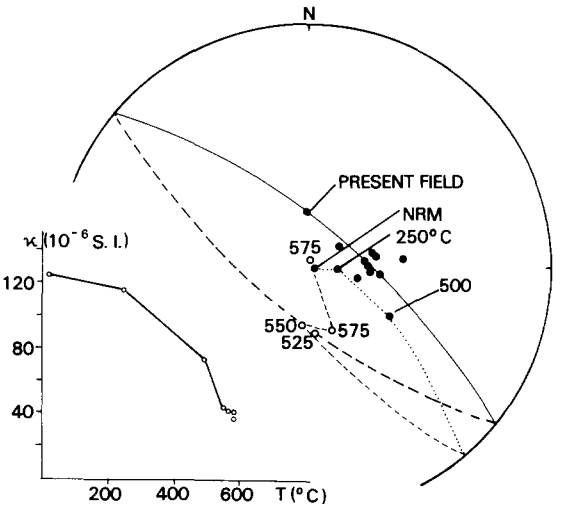


Fig. 5. The variation of susceptibility (left) and remanence direction (right) during demagnetization of a sample from locality R16. The 575 °C step was repeated in the opposite orientation to check the reliability of the measurement at low (5×10^{-5} A m^{-1}) intensities. Also shown in the stereographic plot are the remanence directions of the individual samples from this locality after 500 °C (individual dots) and the remagnetization path on which these directions fall (● = positive inclinations, ○ = negative inclinations).

area the demagnetization experiments alone are not always sufficient to define the characteristic magnetizations unambiguously.

Moreover, there can be problems with the resolution of the magnetic cleaning. Figure 5 shows, among other things, the demagnetization path for a sample from locality R16. The steady decrease of the susceptibility on heating (left figure) indicates the presence of maghemite. The intensity drops remarkably in every sample between 500 and 550 °C and the remanence directions group well after 500 °C (individual dots in right figure). Of the ten samples only a single one happens to have a consistent behaviour above 500 °C (Fig. 5), indicating the existence of an underlying component which, however, is not resolvable in the rest of the samples.

The locality mean directions of magnetization isolated by thermal demagnetization are listed in Tables I and II. The following review of the directions (after tectonic correction) is meant to help select all those that can be retained for further interpretation. It covers each tectonic unit of the study area and each epoch of the Triassic (cf., Tables I and II).

3.1. Aggtelek Mountains

3.1.1. Silica nappe, Early Triassic (Fig. 6a)

There are three normal and one reversed directions. One of the normals (A2) plots significantly on a great circle joining the normal and reversed directions. We take this as a sign of early overprinting and reject the corresponding result (asterisk in Table I).

TABLE III

Summary of palaeomagnetic results from the Aggtelek–Rudabánya Mountains ($\phi = 48.5°$ N, $\lambda = 20.6°$ E)

Geological age epoch	n/N	Mean palaeomagnetic directions								Palaeomagnetic poles			
		Before tectonic tilt correction				After tectonic tilt correction							
		$D°$	$I°$	k	α_{95} (°)	$D°$	$I°$	k	α_{95} (°)	ϕ (°)	λ (°)	dp (°)	dm (°)
Aggtelek Mountains: Silica nappe (s.l.)													
Early Triassic	3/20	294	51	7	49	294	24	31	23	25	281	13	25
Middle Triassic	4/22	294	42	30	17	272	40	34	16	18	305	12	19
Late Triassic	6/45	259	46	12	20	289	59	38	11	40	309	12	16
Rudabánya Mountains: Bódva nappe (s.s.)													
Middle Triassic	5/40	33	72	5	41	298	43	20	17	37	288	13	21

n: Number of useful localities. N: Number of useful samples. Other symbols are self-explanatory. Note: k and α_{95} were computed using n for the number of data points.

3.1.2. Silica nappe, Middle Triassic (Fig. 6b)

There are three normal and one reversed directions which group well.

3.1.3. Silica nappe, Late Triassic (Fig. 6c)

From the five normal and one reversed directions recorded, the latter fits into the cluster of the normals fairly well.

All of these suggest that 13 mean directions of magnetization from the Aggtelek Mountains, which we propose to retain for further analysis, are free of appreciable overprinting and are consistent in showing westerly declinations and decreasing inclinations with age. The overall mean directions are shown in Fig. 6d for each epoch of the Triassic (cf., Table III).

3.2. Rudabánya Mountains

3.2.1. Bódvarákó pseudo-autochthonous beds, Middle Triassic (Fig. 7a)

There are four normal directions displaying a great circle distribution. The smearing of the mean directions indicates that the corresponding magnetizations are still overprinted so that no characteristic remanence direction is derived from these results (asterisk in Table II).

3.2.2. Martonyi nappe, Middle–Late Triassic (Fig. 7b)

Here we have three widely different directions and none of them is meaningful (asterisk in Table II). The negative result can probably be attributed

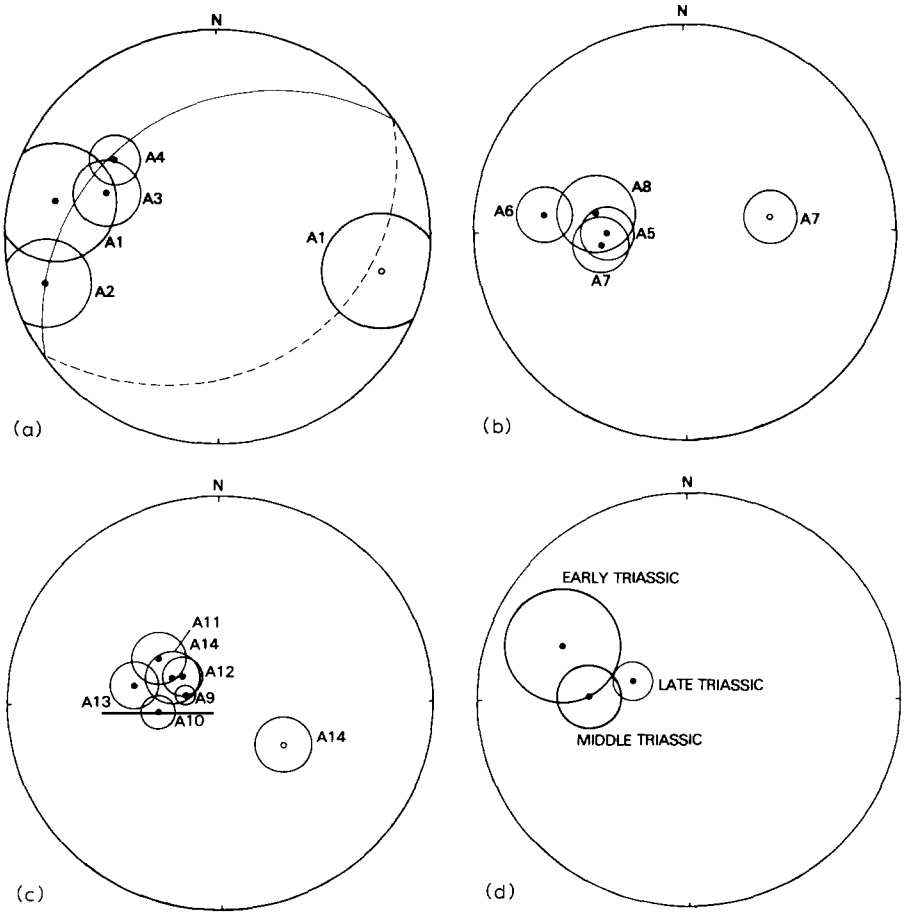


Fig. 6. Stereographic projections of mean remanence directions with α_{95} confidence circles from the Aggtelek Mountains after tectonic correction. Points A1–A14 refer to sampling localities (cf., Table I). ● = positive inclinations, ○ = negative inclinations. (a) Silica nappe (s.s.), Early Triassic, (b) Silica nappe (s.s.), Middle Triassic, (c) Silica nappe (s.l), Late Triassic, (d) mean palaeomagnetic directions from the Aggtelek Mountains for each epoch of the Triassic (cf., Table III).

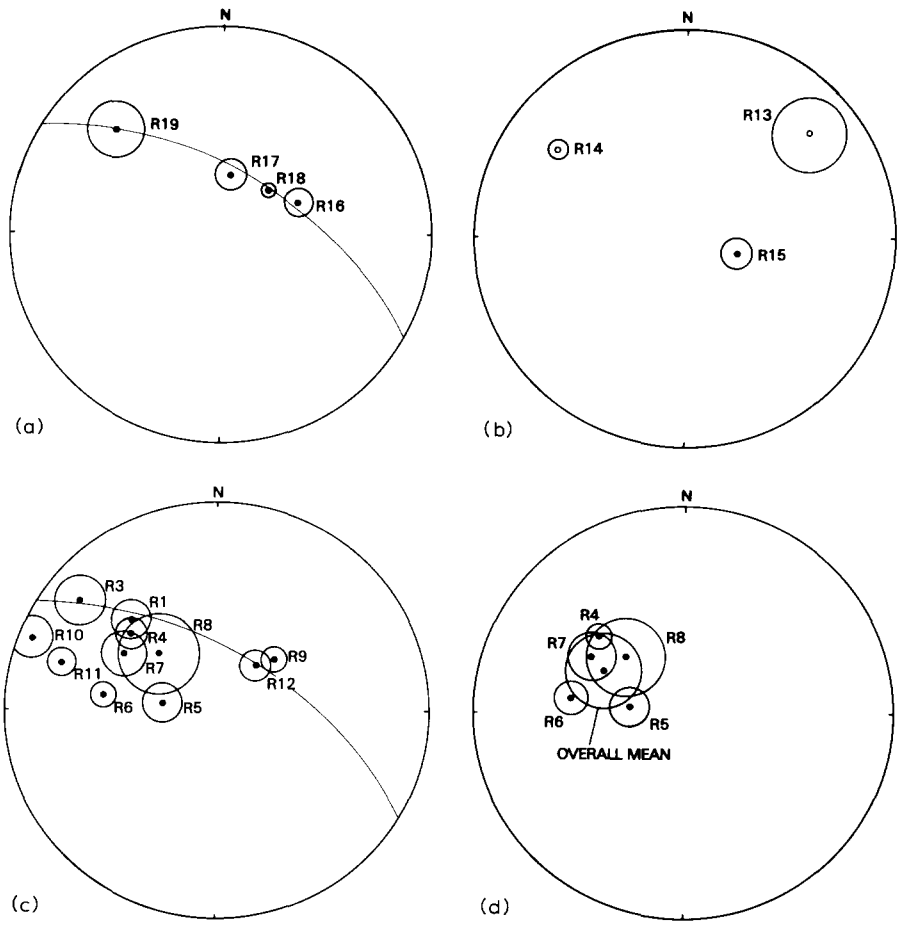


Fig. 7. Mean remanence directions with α_{95} confidence circles from the Rudabánya Mountains after tectonic correction. Stereographic diagrams. Marks R1–R19 refer to sampling localities (cf., Table II). ● = positive inclinations, ○ = negative inclinations. (a) Bódvarákó pseudo-autochthonous beds, Middle Triassic. Note the smeared distribution of the data which is attributed to remagnetization effects. (b) Martonyi nappe, Middle–Late Triassic. (c) Bódva nappe, Middle–Late Triassic. Note that the means R1, R3, R9 and R12 are smeared in a similar way to those in (a). (d) Mean palaeomagnetic directions from the Bódva nappe for the Middle Triassic including the grand mean (cf., Table III).

to a weak metamorphism to which these rocks were subjected.

3.2.3. Bódva nappe, Middle–Late Triassic (Fig. 7c)

From the 11 normal directions, four (R1, R3, R9, R12) fall on the same great circle as the Bódvarákó data (cf., Fig. 7a), hence, overprinting is suspected. Another two (R10, R11) have inclinations too low for their age (Late Triassic), probably due to an unrecognized complexity in the tectonic position of the corresponding sampling sites. With these rejected directions (asterisk in Table II), the remaining five mean directions group well and give an overall mean close to the one for the corresponding Silica unit (Fig. 7d).

Thus we conclude that, from the Rudabánya Mountains, meaningful results can be selected only from the Bódva nappe for the Middle Triassic. The failure of the majority of the Rudabánya rocks to retain the 'primary' magnetization component can probably be attributed to tectonic

deformation stronger than in the Silica nappe and a low-grade metamorphism which is absent in the Silica unit.

4. Discussion and conclusions

Meaningful remanence directions could be isolated from the Silica nappe for all three epochs of the Triassic and from the Bódva nappe for the Middle Triassic only. These are summarized in Tables I and II (entries without asterisks) both before and after tectonic correction with other relevant data. Table III shows the corresponding overall mean directions and the palaeomagnetic poles computed from the overall means after tectonic correction.

Obviously, these results satisfy certain reliability criteria, such as stability against thermal treatment, the presence of both normal and reversed magnetizations with directions significantly different from that of the present geomagnetic field, and positive response to the tectonic fold test. (The reason why unfolding improves the grouping for the Middle Triassic from the Silica nappe only marginally is that the bedding attitudes are similar in these sampling localities.) Thus, we are confident that the retained data represent true palaeomagnetic directions characteristic of their stratigraphical ages. The consistency of the palaeodeclinations as well as the close agreement between the Middle Triassic palaeoinclinations from the Silica and Bódva nappes supports this interpretation of the tectonically corrected remanence directions. The final argument for the primary nature of the palaeomagnetic directions is the trend of the palaeoinclinations, which increase with decreasing age during the Triassic from the low Permian to the high Jurassic inclinations to be expected for these areas.

Figure 8 shows the palaeomagnetic pole positions of Table III together with the mean Triassic palaeomagnetic poles for Europe and Africa (Márton, 1984). Pole positions falling on the broken line (small circle) give identical palaeoinclinations in the sampling area which is marked by a thick arrow. Clearly, the Triassic palaeopoles for both Europe and Africa, as well as the measured

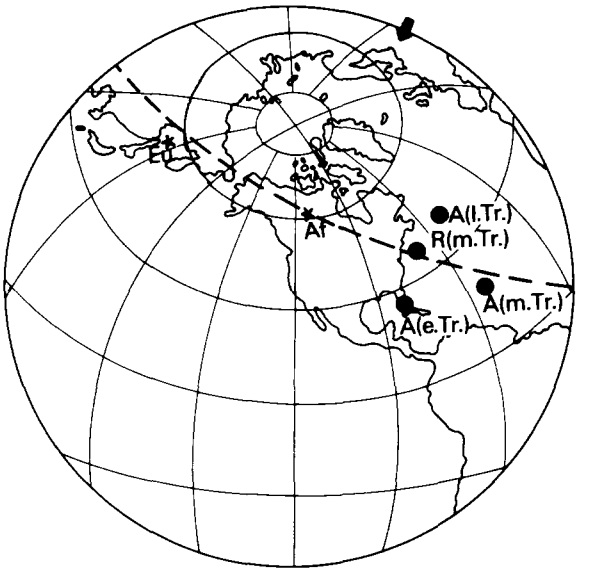


Fig. 8. Palaeomagnetic pole positions from the Aggtelek (A) and Rudabánya (R) Mountains for the Early, Middle and Late Triassic (cf., Table III). The Triassic poles for Europe (Eu) and Africa (Af) are shown for comparison. The small circle indicated by the broken line is centred on the sampling area (thick arrow in the upper right).

ones for the Aggtelek–Rudabánya Mountains, are located along or close to this circle, which indicates that these units may have been part of either continent during the Triassic. The westerly rotation of the palaeodeclinations, however, which on average are 50–60° and 110° relative to Africa and Europe, respectively, suggest an African rather than European affinity for the area. But as our data represent hardly more than a spot-reading of the apparent polar wander path, at present, it is not possible to tell with any certainty from the data alone whether the study area is an 'African' or 'European' unit in the tectonic sense (i.e., moved together with either continent after the Triassic).

In any case, it is certain that the Aggtelek–Rudabánya Mountains have rotated significantly after the Triassic (probably in the Neogene) and the sense of rotation is the same (i.e., anticlockwise) as, for example, for the Transdanubian Central Mountains (Márton and Márton, 1981, 1983), the Slovakian Ore Mountains (Muska and Vozár, 1984) and many other units in the Central Mediterranean area (e.g., VandenBerg and

Zijderveld, 1982). Therefore, we speculate that all westerly declination rotations within this larger area (Márton, 1987) might have a common underlying cause, for which we suggest a more or less uniform system of large-scale tectonic movements that brought these units into their present positions. In the Aggtelek–Rudabánya area neither small-scale relative movements nor nappe transport seems to have affected appreciably the general rotation pattern, at least within the resolution of the data.

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